MRINMOY DATTA

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Basundhara, Flat no. F, 2nd Floor
199/1 and 199/2 Banerjee Para Road
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Current Position

Assistant Professor,
Indian Institute of Technology Hyderabad — July 2020 -

POST-DOCTORAL EXPERIENCES

Postdoctoral Fellow,
University of Tromsø — June 2018 - May 2020
The Arctic University of Norway, Tromsø, Norway
Mentor - Prof. Trygve Johnsen

Postdoctoral Fellow,
Technical University of Denmark, — June 2016 - May 2018
Copenhagen, Denmark
Mentor - Prof. Peter Beelen

Research Assistant,
Indian Institute of Technology Bombay — March 2016 - May 2016

EDUCATION

Doctor of Philosophy, Mathematics
Indian Institute of Technology Bombay — 2016
Thesis Title - Rational points on linear sections of
algebraic varieties over finite fields and higher weights of linear codes.
Thesis Supervisor - Prof. Sudhir R. Ghorpade
Date of Defence: 22 March, 2016

Master of Mathematics — 2011
Indian Statistical Insitute, Bangalore Center

Bachelor of Science, Mathematics Hons. — 2009
St. Xavier's College, Kolkata

PUBLICATIONS

1. (with Sudhir Ghorpade) **Higher weights of affine Grassmann codes and their duals**, *Algorithmic Arithmetic, Geometry, and Coding Theory* (Luminy, France, June 2013), S. Ballet, M. Perret, and A. Zaytsev Eds., *Contemporary Mathematics*, Vol. 637, American Mathematical Society, Providence, RI, 2015, pp. 79-91.
2. (with Sudhir Ghorpade) **On a conjecture of Tsfasman and an inequality of Serre for the number of points on hypersurfaces over finite fields**, Dedicated to M. Tsfasman and S. Vladut on their 60th birthdays, *Moscow Mathematical Journal*, Vol. 15, No. 4 (2015), 715-725.
3. (with Sudhir Ghorpade) **Number of solutions of systems of homogeneous polynomial equations over finite fields**, *Proceedings of the American Mathematical Society*, Vol 145, No 2 (2017), 525-541.
4. (with Sudhir Ghorpade) **Remarks on Tsfasman-Boguslavsky conjecture and higher weights of projective Reed-Muller codes**, *Arithmetic, Geometry, Cryptography and Coding Theory (Luminy, France, May 2015)*, A. Bassa, A. Couvreur and D. Kohel Eds., *Contemporary Mathematics*, Vol. 686, American Mathematical Society, Providence, RI, 2017, pp. 157-169.
5. (with Peter Beelen and Sudhir Ghorpade) **Maximum number of common zeros of homogeneous polynomials over finite fields**, *Proceedings of the American Mathematical Society*, Vol 146 No. 4 (2018), 1451-1468.
6. (with Elise Barelli, Peter Beelen, Vincent Neiger, and Johan Rosenkilde)**Two-Point Codes for the Generalized GK curve**, *IEEE Transactions on Information Theory* Vol. 64, No. 9 (2018), 6268-6276.
7. (with Peter Beelen) **Generalized Hamming weights of affine cartesian codes**. *Finite Fields and their Applications* Vol 51 (2018), 130-145.
8. (with Peter Beelen and Sudhir Ghorpade) **Vanishing ideals of projective spaces over finite fields and a projective footprint bound**. *Acta Mathematica Sinica (English Series)* Vol 35 (2019), No. 1, 47-63.
9. **Maximum number of $\mathbb{F}_q$-rational points on nonsingular threefolds in $\mathbb{P}^4$.** *Finite Fields and their Applications.* Vol 59 (2019), 86-96.
10. (with Peter Beelen) **Maximum number of points on intersection of a cubic surface and a non-degenerate Hermitian surface.** *Moscow Mathematical Journal,* Vol 20, No. 3 (2020),, 453–474.
11. **Relative generalized Hamming weights of affine Cartesian codes.** *Designs, Codes and Cryptography*, Vol. 88, No. 6, (2020), 1273–1284.
12. (With Peter Beelen and Masaaki Homma) **A proof of Sørensen's conjecture on Hermitian surfaces.** *Proceedings of the American Mathematical Society*, Vol. 149, No. 4, (2021), 1431–1441.
13. (with Peter Beelen and Sudhir Ghorpade) **A combinatorial approach to the number of solutions of systems of homogeneous polynomial equations over finite fields**, *Moscow Mathematical Journal*, Vol. 22 No.4 (2022), 565–593.
14. (wth Subrata Manna) **A generalization of Gerzon's bound on spherical s-distance sets**. *Periodica Mathematica Hungarica* Vol. 87, No. 1, (2023), 51–56.
15. (with Trygve Johnsen) **Codes from symmetric polynomials**, *Designs, Codes, and Cryptography* Vol. 91 (2023), 747–761.

16. (with Subrata Manna) **Maximum number of points on an intersection of a cubic threefold and a non-degenerate Hermitian threefold**, *Finite Fields and Their Applications*, Vol. 98 (2024), 102462.
17. (with Peter Beelen, Maria Montanucci, and Jonathan Niemann) **Intersection of irreducible curves and the Hermitian curve**, *Journal of Algebra*, Vol. 671 (2025), 75 - 94.
18. **Remarks on second and third weights of projective Reed–Muller codes**, *Journal of Algebra and Its Applications*, https://doi.org/10.1142/S0219498825410051.

Preprint

1. (with Tiasa Dutta) **Second minimum weight of Grassmann codes**, arXiv:2508.18844.

RESEARCH GRANTS

1. **SERB Start-up Research Grant.**

   Title of the project: Bounds on the number of rational points on varieties over finite fields and applications to linear error-correcting codes.

   Role: Principal Investigator.

   Amount: INR 11,00,827/-.

2. **DST-RSF Research Grant**

   Title of the project: Classical and Quantum Error-Correcting Codes and Mathematics over Finite Fields for Smart Telecommunications.

   Role: Co-Principal Investigator.

3. **NBHM Research Grant**

   Title of the project: On the number of points on certain varieties over finite fields with applications to coding theory.

   Role: Principal Investigator.

   Total Amount: INR 3,53,000/.

ORGANIZATIONAL ACTIVITIES

1. Co-organizer (with Prof. P. A. Lakshminaraya), One Day Symposium on Algebra and Number Theory, December 23, 2001.
2. Organizer of NCM-CIMPA School on Finite Geometry and Coding Theory, IIT Hyderabad, November 20 - December 1, 2023.
3. Organizer of International Conference on Algebraic Geometry, Coding Theory, and Combinatorics (in honor of Prof. Sudhir Ghorpade's 60th birthday), IIT Hyderabad, December 4 - 8, 2023.

ACADEMIC ACHIEVEMENTS

1. Ranked 1st in B.Sc. (Mathematics Hons.), St. Xavier's College, Kolkata.
2. Recipient of NBHM (National Board for Higher Mathematics) M.Sc. Scholarship 2010 - 2011.
3. Recipient of NBHM Ph.D. Scholarship, 2011 - 2016.
4. Qualified National Eligibility Test (NET) conducted by UGC-CSIR, June 2011 with All India Rank 34.

5. Recipient of NBHM Travel Grant for attending the CIMPA School on *Algebraic Curves over Finite Fields and Applications*, held at the University of Philippines Diliman, Manila, Philippines, July 22 - August 2, 2013. (Received support from CIMPA towards boarding and lodging.)
6. Recipient of *NANUM travel grant for 1000 Mathematicians* for attending International Congress of Mathematicians 2014, held in Seoul, Korea during August 13 - 21, 2014.
7. Recipient of Post-Doctoral Fellowship from Danish Council of Independent Research.
8. Selected for INSPIRE Faculty Award, 2017 (Session II), not availed.
9. Nominated as a member of CIMPA (Center International de Mathématiques Pures et Appliquées).
10. Recipient of travel grant from IMU to attend ICM 2026.

Ph.D. STUDENT SUPERVISION

- Mr. Subrata Manna (Completed).
- Ms. Tiasa Dutta (Ongoing).
- Ms. Bhargavi Dikshit (Ongoing).

REFERENCES

1. Prof. Sudhir Ghorpade
   Department of Mathematics,
   IIT Bombay,
   Email: sudhirghorpade@gmail.com.
2. Prof. Peter Beelen
   Department of Applied Mathematics and Computer Science,
   Technical University of Denmark,
   Email: pabe@dtu.dk
3. Prof. Trygve Johnsen
   Institute of Mathematics and Statistics,
   University of Tromsø,
   The Arctic University of Norway,
   Email: trygve.johnsen@uit.no.